Diabetic Retinopathy: Traditional vs Modern Imaging

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I. INTRODUCTION

Diabetic retinopathy (DR) is a progressive eye disease caused by diabetes and is one of the leading causes of vision loss among the working-age population worldwide [1]. Early detection of DR is critical for timely intervention and prevention of blindness [2]. Traditional DR screening relies on manual examination of fundus images which is time-consuming, costly, and highly dependent on expert evaluation [3]. To overcome these challenges, many studies have focused on applying machine learning techniques to automatically detect DR. This survey examines recent developments in DR detection using 2D and 3D imaging, highlighting how different modalities and model architectures contribute to diagnostic accuracy. Traditional 2D methods are compared with modern 3D approaches to assess which strategies are most effective for clinical DR screening.

I. TRADITIONAL IMAGING

The most common approach for detecting diabetic retinopathy is based on 2D fundus images, due to their wide-availability and cost-effectiveness. Convolutional neural networks (CNNs) have become the standard method for detecting and grading DR based on standardized severity scales [1]. The CNNs are typically trained on a large dataset of labeled images, and many of them rely on transfer learning techniques since there is a limited number of annotated medical data. Pretrained models such as ResNet, InceptionV3, EfficientNet, and VGG16 are widely used for transfer learning [1]. While many recent models rely on transfer learning; foundational studies also demonstrated the potential of training deep learning models entirely from scratch using large medical datasets, by achieving high sensitivity and specificity [2].

To improve the performance of conventional CNNs in detecting diabetic retinopathy, later studies focused on hybrid architectures that combined different architectures with CNNs. Capsule networks, for example, are combined with CNNs to better capture spatial relationships between retinal features [3]. While CNNs are good at detecting local patterns, they may lose spatial details due to pooling layers [3]. Capsule networks preserve the hierarchy between these features, which helps the model better understand the structure of the retina [3]. By combining the strengths of architectures like capsule networks, these hybrid models provide more detailed information about retinal structures. Therefore, they are more effective than conventional CNNs for detecting diabetic retinopathy.

Despite the improvements achieved by hybrid models, most CNN-based classifiers process fundus images as a whole, which may lead to overlooking clinically significant features. Since diabetic retinopathy begins with early lesions such as microaneurysms, intraretinal hemorrhages, or exudates [4], models that include lesion-level information can provide more clinically meaningful predictions. To address this issue, lesion-aware frameworks have been developed. These models first segment lesions using architectures such as U-Net and then combine the lesion maps with the original fundus image. This combined input allows the model to focus on both the overall image and lesion-specific features, improving performance [4].

To further improve the lesion-focused modeling, recent approaches introduced integrating textual descriptions along with images to help models focus on specific lesion types more effectively. Vision-language frameworks use textual descriptions to guide segmentation, helping the model focus on medically relevant features. For example, models that adapted the Segment Anything Model (SAM) embed lesion descriptions such as “tiny red dots” to better identify the corresponding retinal features [5]. This method demonstrates improved segmentation accuracy, and also helps make the model’s decisions easier for clinicians to interpret [5].

Since all previously discussed methods require annotation for data, several studies focused on reducing the need for labels in detection of DR. Self-supervised and contrastive learning methods allow models to train by comparing different parts

of an image or different images. For example, the model can learn to tell the difference between regions with and without lesions [6]. These methods allow extracting meaningful features when annotated medical data is costly to obtain.

II. MODERN IMAGING

2D fundus imaging remains standard in many clinics limited to surface level visualization and could miss early pathological signs of DR in deeper retinal layers [7]. Optical coherence tomography (OCT) and optical coherence tomography angiography (OCTA) offer depth resolved, noninvasive imaging of retinal structure and vasculature making them highly valuable for detecting early and subtle changes [8]. However, these technologies produce complex 3D volumes that are difficult to interpret manually and prone to motion artifacts [9]. As a result, recent studies have focused on improving quality, interpretability and automation of 3D imaging to enable early, accurate and scalable DR screening.

One persistent obstacle in 3D imaging is motion artifact which is especially seen in patients with poor fixation and common among those with vision threatening DR. Recent work has introduced software pipelines that register and average multiple OCTA volumes to address this problem [9]. These techniques remove the need for a stable reference image or hardware stabilization while reconstructing clearer volumetric renderings [9]. Importantly, they reveal pathologies in deeper capillary plexuses and choriocapillaris that are often invisible in conventional 2D slices [9].

Focus of challenge shifted to interpreting high resolution 3D volumes once motion artifacts are mitigated. One strategy is to directly classify volumetric OCT and OCTA data using 3D CNNs. Model uses a semi sequential 3D-CNN architecture that processes both structural OCT and angiographic OCTA volumes. Allow network capture spatial and vascular features simultaneously [7]. Also it generates 3D CAMs which highlights relevant retinal regions for diagnostic decisions [7]. These models learn specific features of disease from raw inputs. Some frameworks can even classify DR, AMD and glaucoma simultaneously. They are facilitating scalable multi disease screening [7].

An alternative relies on extracting interpretable structural features like retinal layer reflectivity and thickness from segmented 3D scans. Method extracts voxel level reflectivity and thickness features from segmented retinal layers. Aggregates them into 3D structural maps which are used as input to a neural network for doing classification [10]. This feature extraction increases interpretability and decreases computational load compared to end to end deep models [10]. These features are derived from known DR biomarkers then fed into neural networks for classification. This method trades end to end automation for greater interpretability and lower computational load and making it suitable for clinics that have limited infrastructure [10].

Another focus is on segmentation of fluid accumulation in diabetic macular edema which is a primary cause of vision loss in DR. Volumetric segmentation of retinal fluid is more informative than central thickness measures or area estimates. ReF-Net is based on a modified U-Net architecture that integrates structural channels to accurately segment retinal fluid. Network distinguishes fluid from vasculature by incorporating vessel aware features [8]. These systems improve differentiation between fluid and vascular tissue by incorporating angiographic signals [8]. Accurately producing artifact resistant fluid maps that can guide treatment decisions. Across all approaches machine learning models are optimized using standard loss functions such as cross entropy for classification and dice coefficient for segmentation and are trained using annotated OCT/OCTA volumes to learn spatial, structural and angiographic patterns indicative of DR [8].

In this survey we compared 2D and 3D imaging methods that detect DR. 2D fundus images are still widely used but they often miss early signs in the deeper layers of the retina. This is where 3D methods such as OCT and OCTA come to help and show more detail. Machine learning modes made it possible to use complex images without need for any expert review. Studies have shown deep learning can solve problems like motion artifacts, improve classification and detect fluid in the retina. These developments make 3D imaging more useful in clinics. It also allows for earlier and more accurate diagnosis helping to prevent vision loss.

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